Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

DEPARTMENT OF MECHATRONICS
VII SEMESTER B.TECH. MECHATRONICS

END SEMESTER EXAMINATIONS, DEC. 2021

SUBJECT: ENGINEERING MATERIALS [MTE 4071]

Date: 20/12/2021

MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Data not provided any, may be assumed suitably.

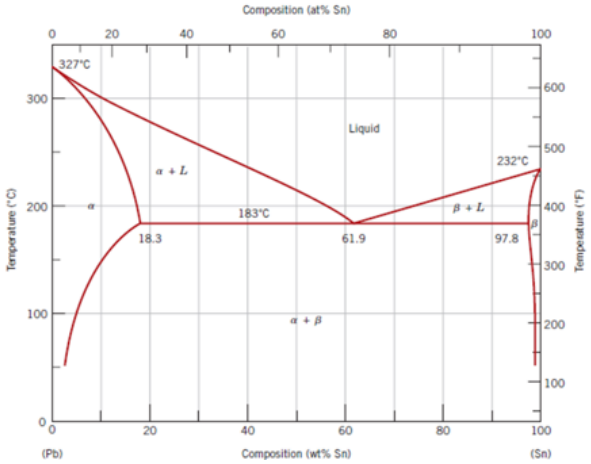
Q. No	PART – A (30×1 = 30 Marks) 50 Mins.	M	CO	PO	LO	BL
1.	Forces applied to a segment of material lead, to an appearance of electrical charge on the surfaces of the segment is called as _____ **a. Piezoelectric Effect** b. Magnetostrictive Effect c. Conduction d. Magnetization	1	5	4	4	2
2.	Generation of stress when an electric field is applied is called as _____ a. Direct Piezoelectric Effect b. **Converse Piezoelectric Effect** c. Semi conductivity d. Two-way shape memory effect	1	5	4	4	2
3.	Which among the following is not a natural piezoelectric material? a. Berlinite b. Quartz c. Tourmaline **d. Barium Titanate**	1	5	4	4	2
4.	Materials developing large mechanical deformations when subjected to an external magnetic field are called as _____ a. Piezoelectric Materials b. Semiconducting Materials **c. Magnetostrictive Materials** d. Optical Materials	1	5	4	4	2

5.	Change in Magnetization due to applied stress is called as _____ **a. Villari Effect** b. Joule Effect c. Nagoka-Honda Effect d. Wiedemann effect	**1**	**5**	**4**	**4**	**2**
6.	Torque induced by helical magnetic field is called as _____ a. Villari Effect b. Joule Effect c. Nagoka-Honda Effect d. **Wiedemann effect**	**1**	**5**	**4**	**4**	**2**
7.	Change in the magnetic state due to change in the volume is called as _____ a. Villari Effect b. Joule Effect c. **Nagoka-Honda Effect** d. Wiedemann effect	**1**	**5**	**4**	**4**	**2**
8.	Austenite phase in a shape memory alloy (SMA) is characterized by a. Simple FCC structure b. High Temperature state c. Hard, firm d. **All of the above**	**1**	**5**	**4**	**4**	**2**
9.	Example for magnetic material used in data storage devices _____ a. 45 Permalloy **b. CrO_2** c. Cunife d. Alnico	**1**	**5**	**4**	**4**	**2**
10.	The effective number of lattice points in the unit cell of simple cubic, body centered cubic, face centered cubic space lattices are ___ a. 1,2,2 **b. 1,2,4** c. 2,3,4 d. 2,4,4	**1**	**1**	**4**	**4**	**2**
11.	If a particular Fe-C alloy contains less than 0.83% carbon, it is called as____ a. High Speed Steel b. Hypoeutectoid Steel **c. Hypereutectoid Steel** d. Cast Iron	**1**	**1**	**4**	**4**	**2**

12.	Which one of the following factors is more relevant to represent complete solubility of two metals in each other? a. Chemical Affinity b. Valency Factor c. Crystal Structure Factor **d. Relative Size Factor**	**1**	**4**	**4**	**4**	**2**
13.	The correct statement for Burger's vector in screw dislocation is _____ a. Perpendicular to dislocation line b. Inclined to dislocation line **c. Parallel to dislocation line** d. Opposite to dislocation line	**1**	**3**	**4**	**4**	**2**
14.	Pearlite phase in an iron-carbide diagram is ___ a. Eutectic Phase b. Hypoeutectic Mixture **c. Eutectoidal Mixture** d. Hypereutectic Mixture	**1**	**4**	**4**	**4**	**2**
15.	Austempering is employed to obtain ___ a. 100% Marternsite **b. 100% Bainite** c. 50% Martensite and 50% Bainite d. 100% Pearlite	**1**	**4**	**4**	**4**	**2**
16.	In a conventional annealing process, austenite transforms into ____ **a. Coarse Pearlite** b. Fine Pearlite c. Upper Bainite d. Medium Pearlite	**1**	**4**	**4**	**4**	**2**
17.	If a steel alloy having either pearlitic or bainitic microstructures is heated to, and left at, a temperature below the eutectoid for a sufficiently long period of time, the resulting microstructure formed is ____ a. Martensite b. Tempered Martensite **c. Spherodite** d. Austenite	**1**	**4**	**4**	**4**	**2**

18.	Alternative layers of α – ferrite and Fe_3C that are relatively thin is observed in ___ a. Bainite **b. Fine Pearlite** c. Coarse Pearlite d. Martensite	**1**	**4**	**4**	**4**	**2**
19.	A cube elongated in one direction such that a=b≠c is called as ___ **a. Tetragonal Structure** b. Orthorhombic Structure c. Monoclinic d. Triclinic	**1**	**1**	**4**	**4**	**2**
20.	The magnitude of the electric field required to cause dielectric breakdown is called as ____ **a. Breakdown Strength** b. Dielectric Loss c. Polarization d. Space Charge	**1**	**5**	**4**	**4**	**2**
21.	If the magnetic susceptibility of a material is small, constant and negative, the magnetism shown by the material is ____ a. Paramagnetism b. Ferromagnetism **c. Diamagnetism** d. Ferrimagnetism	**1**	**5**	**4**	**4**	**2**
22.	Highest specific modulus and highest specific strength is observed in ___ a. Aramid Fibre-reinforced polymer composites **b. Carbon Fibre-reinforced polymer composites** c. Glass Fibre-reinforced polymer composites d. Metal Matrix Composites	**1**	**5**	**4**	**4**	**2**
23.	In an AA6061 metal matrix, boron as reinforcement material, the fiber content is ___ **a. 48 vol.%** b. 45 vol.% c. 52 vol.% d. 38 vol.%	**1**	**5**	**4**	**4**	**2**
24.	A resolution range of 10^{-10} to 10^{-4} is observed in ____ a. Scanning Probe Microscope b. Scanning Electron Microscope **c. Transmission Electron Microscope**	**1**	**3**	**4**	**4**	**2**

	d. Optical Microscope					
25.	The percentage of carbon at eutectic point in Fe-C phase diagram is ____ **a. 4.3** b. 2.1 c. 0.76 d. 0.2	**1**	**4**	**4**	**4**	**2**
26.	Mechanical properties of fiber-reinforced composites depend on _____ a. Properties of Constituents b. Interface Strength c. Fibre length, orientation and volume fraction **d. All of the above**	**1**	**4**	**4**	**4**	**2**
27.	If a Fe-C alloy is cooled at a rate greater rate than that of CCR, the product formed is ___ a. Bainite **b. Martensite** c. Pearlite d. Austenite	**1**	**4**	**4**	**4**	**2**
28.	The heat treatment process to enhance the ductility and toughness of martensite, relieving internal stresses is done by ____ a. Annealing b. Normalizing **c. Tempering** d. Austempering	**1**	**4**	**4**	**4**	**2**
29.	The attribute of a material which resists the flow of electricity is known as ____ a. Conductivity b. Thermoelectricity c. Dielectric strength **d. Resistivity**	**1**	**5**	**4**	**4**	**2**
30.	Pyrometer works based on ___ a. Laser technology b. Photo-conduction **c. Thermal emission** d. Tyndall effect	**1**	**5**	**4**	**4**	**2**

PART – B (20 MARKS)						75 Mins.
1A.	For the 40 wt.% Sn – 60 wt.% Pb alloy, calculate the relative amount of each phase present in terms of: (a) mass fraction, and (b) volume fraction. At 150°C, take the densities of Pb and Sn to be 11.23 and 7.24 g/cm³, respectively. Refer figure 1A.  ***Figure 1A***	**4**	**4**	**2**	**2**	**3,4**
1B.	The metal niobium has a BCC crystal structure. If the angle of diffraction for the (211) set of planes occurs at 75.99° (first-order reflection) when monochromatic x-radiation having a wavelength of 0.1659 nm is used, Compute: **(a)** the interplanar spacing for this set of planes, and **(b)** the atomic radius for the niobium atom.	**3**	**2**	**2**	**1**	**3**
1C.	Mr. X fabricated an XX sample for his research purposes. He wants to do characteristic study the topography of the specimen on an atomic scale and with high magnification in the range of 10^9×. As XX is reactive in normal conditions, the study has to be done in vacuum environment. Assess Mr. X's requirements and suggest a suitable examination technique. Discuss its working principle.	**3**	**3**	**4**	**4**	**4**
2A.	The magnetization within a bar of some metal alloy is 1.2×10^6 A/m an H field of 200 A/m. Calculate the following: **(a)** magnetic susceptibility, **(b)** permeability, and **(c)** magnetic flux density within this material. For the above-mentioned alloy, infer what type of magnetism is exhibited by the alloy and Why?	**4**	**5**	**2**	**1**	**3,4**

	Permeability of vacuum is 1.257×10^{-6} H/m.					
2B.	Explain how piezoelectric effect affects the sensing and actuating property in a piezo pressure sensor.	**3**	**5**	**4**	**4**	**2**
2C.	Discuss how magnetostrictive sensor is used in hydraulics/pneumatic cylinders to determine the end stroke and home of the piston head of the cylinder.	**3**	**5**	**4**	**4**	**2**